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Camera Control PTP 2 Reference

For Models Earlier than 2020

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Document History

Date	Version	Description
Nov. 1 2023	1.0.0	• First version
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Contents

Overview	7
Connect	8
Get Property	9
Set Property	10
Send Control Command	11
Get Image File	12
Live View	14
Movie Recording	14
Disconnect	14
Command List	15
Operations	15
Device Properties	15
Controls	16
Events	17
Vendor Response Code	17
Object Format	17
Compatibility	19
Operations	19
Device Properties	19
Controls	20
Events	20
Operations	21
GetDeviceInfo	21
OpenSession	21
CloseSession	21
GetObject	21
GetThumb	22
GetPartialObject	22
SDIO_Connect	23
SDIO_GetExtDeviceInfo	23
SDIO_SetExtDevicePropValue	25
SDIO_ControlDevice	25
SDIO_GetAllExtDevicePropInfo	26
Device Properties	29
Compression Setting	29
White Balance	30
F-Number	31
Focus Mode	33
Exposure Metering Mode	34
Flash Mode	35
Exposure Program Mode	36
Exposure Bias Compensation	38

Still Capture Mode	38
Flash Compensation	43
Dynamic Range Optimizer	44
Image Size	45
Shutter Speed	46
Battery Level Indicator	47
Color Temperature	48
Biaxial Fine-Tuning G-M Direction	48
Aspect Ratio	49
Focus Indication	50
Predicted Maximum File Size	50
Shooting File Info	51
AELock Indication	52
Battery Remaining	52
Picture Effect	53
Biaxial Fine-Tuning A-B Direction	54
Movie Recording State	55
ISO Sensitivity	55
FELock Indication	60
Live View Status	60
Still Image Save Destination	61

Controls62

Shutter Half-Release (S1) Button	62
Shutter Release (S2) Button	62
AEL Button	63
AFL Button	63
Release Lock	64
RequestOneShooting	64
Movie Rec Button (Hold)	65
FEL Button	65
Media Format	66
Focus Magnifier	67
Focus Magnifier Cancel	67
Remote Key UP	68
Remote Key DOWN	68
Remote Key LEFT	69
Remote Key RIGHT	70
AF/MF Hold	70
Cancel Pixel Shift Shooting	71
Pixel Shift Shooting Mode	72
HFR Standby	72
HFR Recording cancel	73
Focus Step Near	73
Focus Step Far	74
AWBL Button	75

Events76

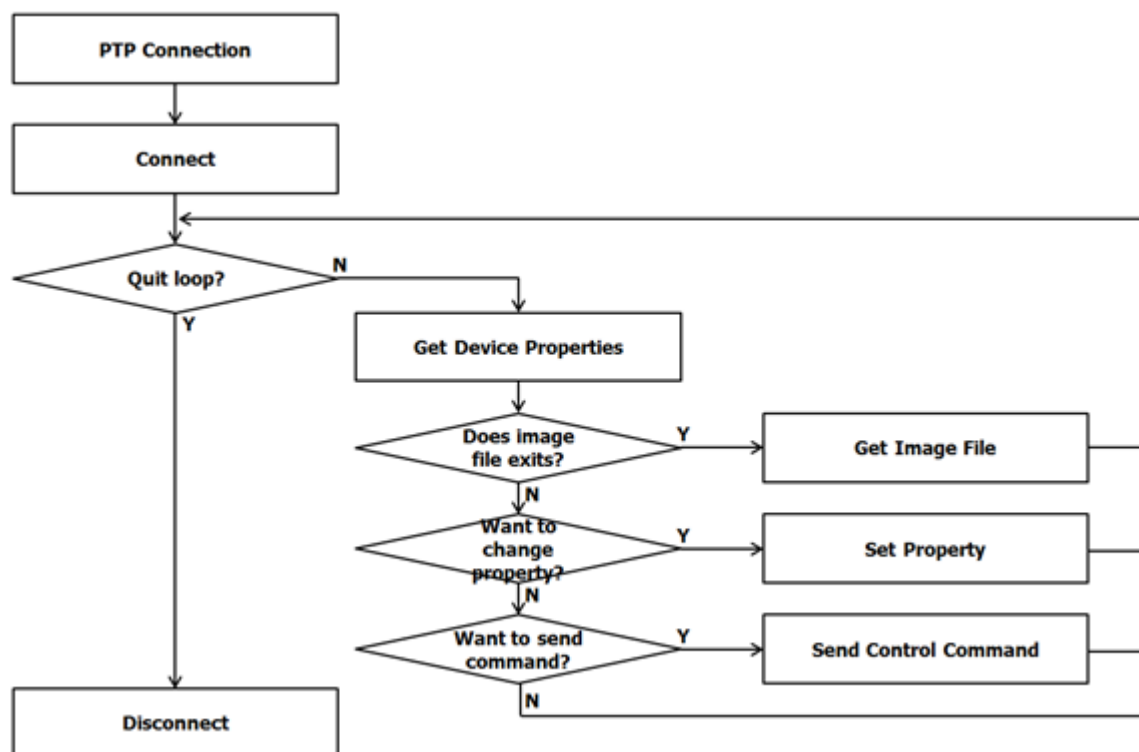
SDIE_ObjectAdded	76
SDIE_ObjectRemoved	76
SDIE_DevicePropChanged	76

More Information78
Trademarks and Acknowledgements 78

Overview

The “Camera Control PTP” is based on PTP (Picture Transfer Protocol) and MTP (Media Transfer Protocol). It uses the PTP vendor extended commands. All vendor-extended operations are prefixed with **SDIO_** and events with **SDIE_** in this document. This document lists the command specifications and explains how to use the “Camera Control PTP” function. Please refer to the PIMA 15740 standard, the USB Still Image Capture Device definition, and the Media Transfer Protocol Spec document for further explanations on these protocols.

The “Camera Control PTP” sequence consists of the following parts: connect, get device properties, get image files, set device properties, and send control commands. The protocol is described in detail in the following pages. Here is an overall sequence.



In this communication sequence, the Initiator, such as the PC application software, should not use PTP vendor events because earlier camera models do not guarantee that all events are delivered. The Initiator must periodically obtain the properties (status) of the Responder, that is, the camera, if the Responder and Initiator are not busy, such as when transferring image files. Events are suitable if the application only targets recent camera models (2019 or later). In this case, ensure that your application constantly receives all the USB queue buffers; otherwise, some events may be dropped.

To obtain these properties, use [SDIO_GetAllExtDevicePropInfo](#). The property structure includes all available camera statuses (such as shutter speed, F-number, ISO), information about queued images in the transfer buffer in the camera, and each property’s enabled/disabled status.

After obtaining the properties, the Initiator should check whether the image file exists. If it exists, the Initiator should obtain it immediately using the PTP standard operation [GetObject](#).

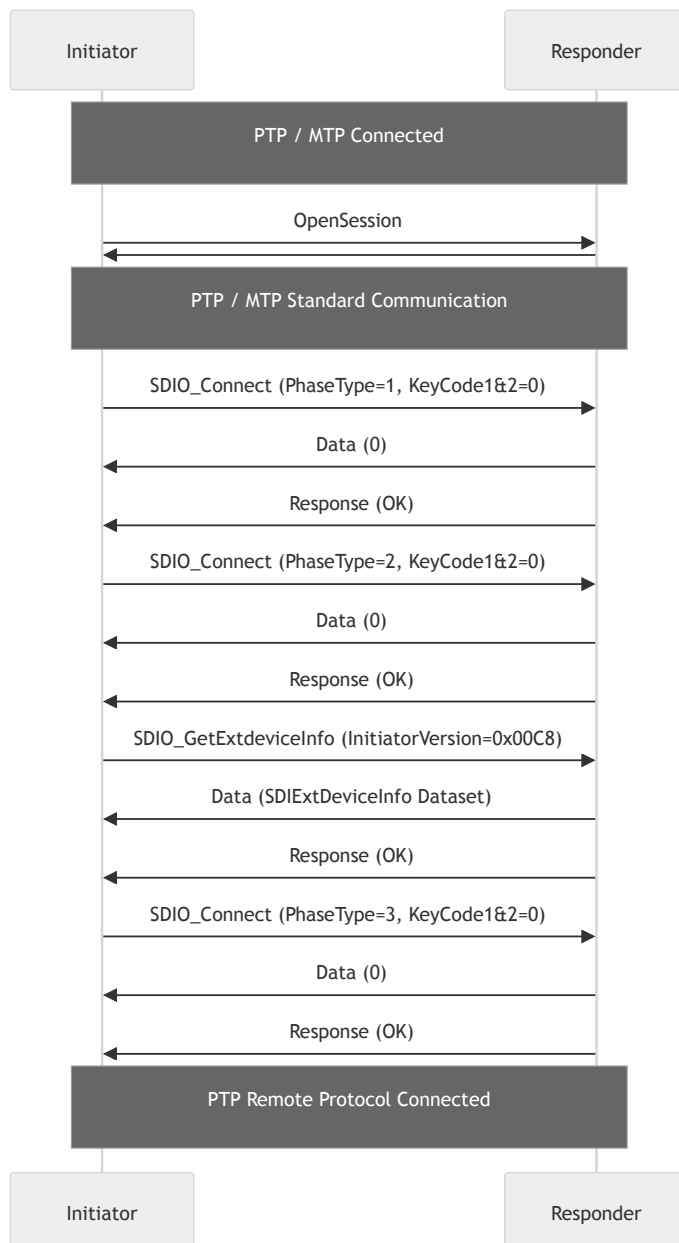
The Initiator can change some property values in the camera by sending a set property command, [SDIO_SetExtDevicePropValue](#).

To capture the picture, start recording the movie, lock AE or AF, and the Initiator sends [Controls](#) to the camera with [SDIO_ControlDevice](#).

The protocol is described in detail in the following section.

Connect

Authentication Sequence:



After the connection between the Initiator and the camera has been established, the Initiator must initiate a connection protocol using [SDIO_Connect](#) and [SDIO_GetExtDeviceInfo](#).

The Initiator must send [SDIO_Connect](#) twice at the beginning.

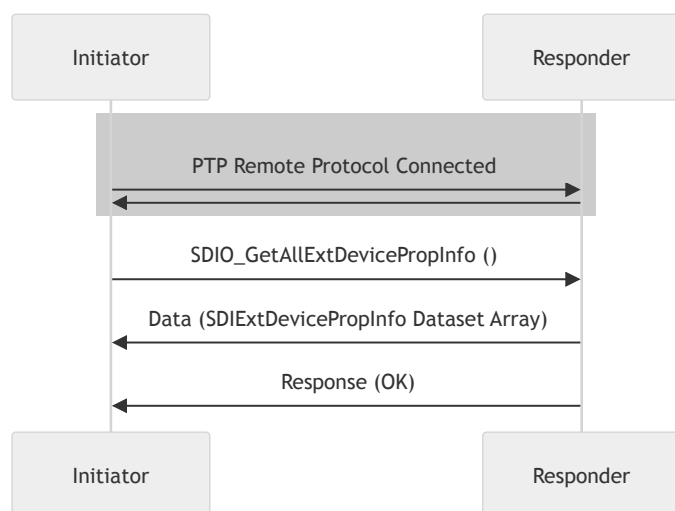
In the first call, `PhaseType` should be `0x01` and `0x02` in the second. Subsequently, [SDIO_GetExtDeviceInfo](#) should be sent. The Initiator can obtain the protocol version using the response code; if the version is okay, call [SDIO_Connect](#) again and establish the connection.

When [SDIO_GetExtDeviceInfo](#) fails (returned data size is zero), retry until successful. After succeeding, [SDIO_Connect](#) must be executed with `PhaseType=0x03`.

Camera Control PTP has its version defined by its specifying functions. This version will change if a supporting function in any command or property changes. The Initiator can check the “Extension Version” in the response code of [SDIO_GetExtDeviceInfo](#) and confirm it is equal to `0x00C8`. Camera functionalities vary among firmware. The Initiator should check the array of properties and control codes.

Get Property

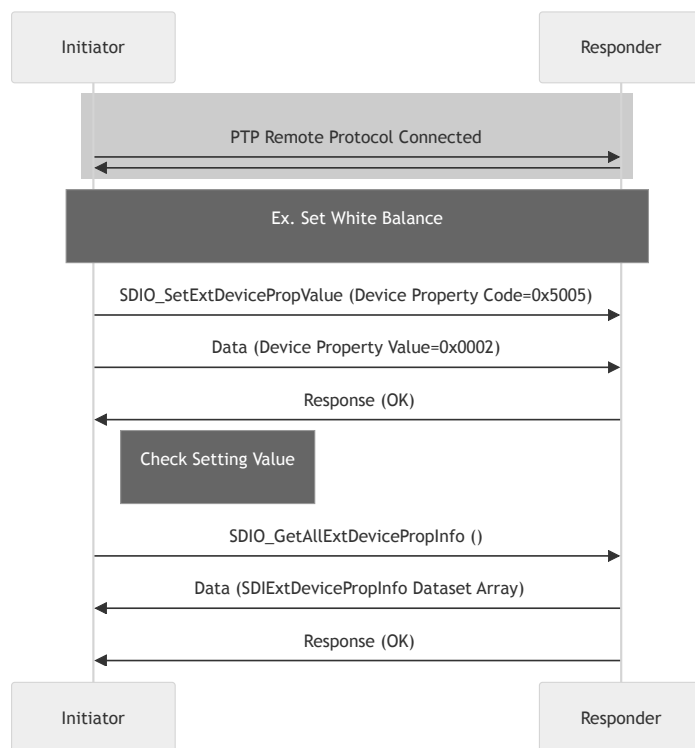
Get Camera Setting Sequence:



After connection to the camera, send [SDIO_GetAllExtDevicePropInfo](#) to obtain the camera status, the list of available properties, and the range of values for each property. The host shall parse the [SDIExtDevicePropInfo Dataset](#) Array returned by the camera and search for the desired property.

Set Property

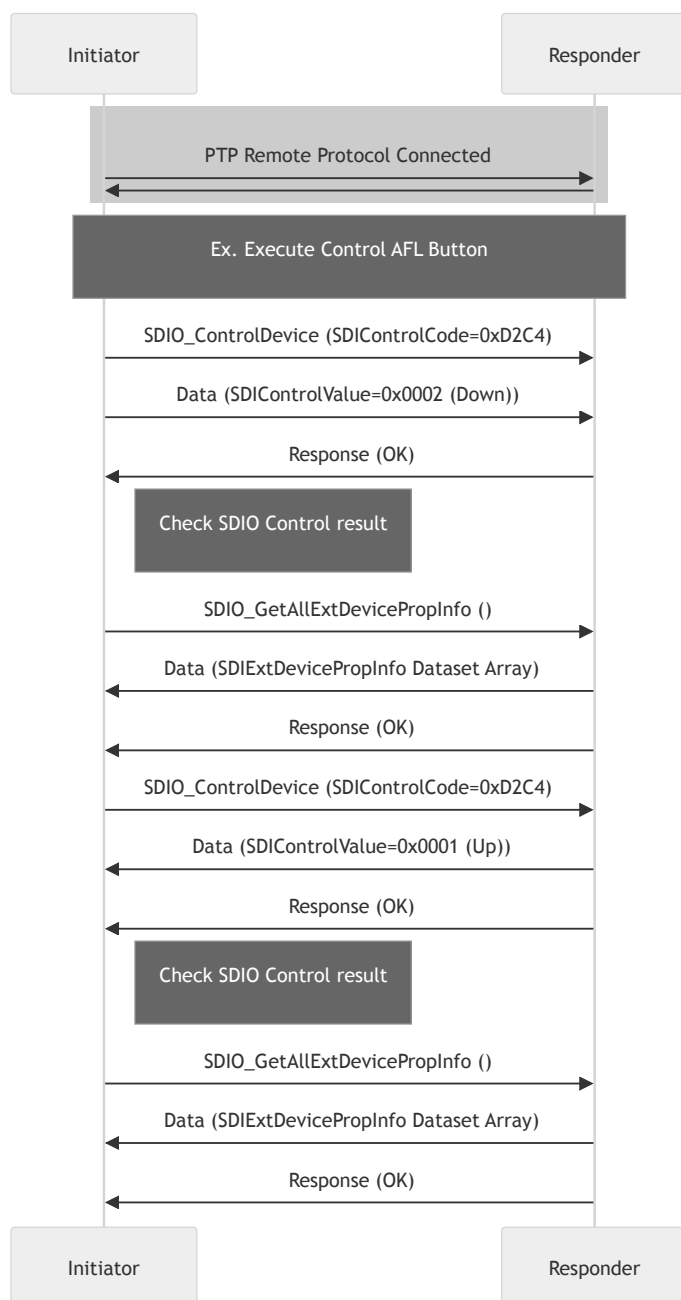
Set Camera Setting Sequence:



The Initiator can set a property value by [SDIO_SetExtDevicePropValue](#).

Send Control Command

Execute SDIO Control Code Sequence:



The Initiator can send a control command to a camera using [SDIO_ControlDevice](#). To know all supported control commands, see the [Controls](#) section in the [Command List](#) chapter. Buttons such as shutter half-release (S1) and shutter release (S2) have an on/off status, whereas controls such as F-number and shutter speed have an up/down status with step value as a control value.

S1 indicates the half-pressing shutter release button, and S2 is the full-pressing shutter release button. Therefore, the Initiator can control the interval from push to release using the S2 on/off. But this should emulate the camera's hardware operation; the Initiator should call in proper order like S1 on → S2 on → S2 off → S1 off. If the Initiator does not send data in the proper order, the camera's behavior is not

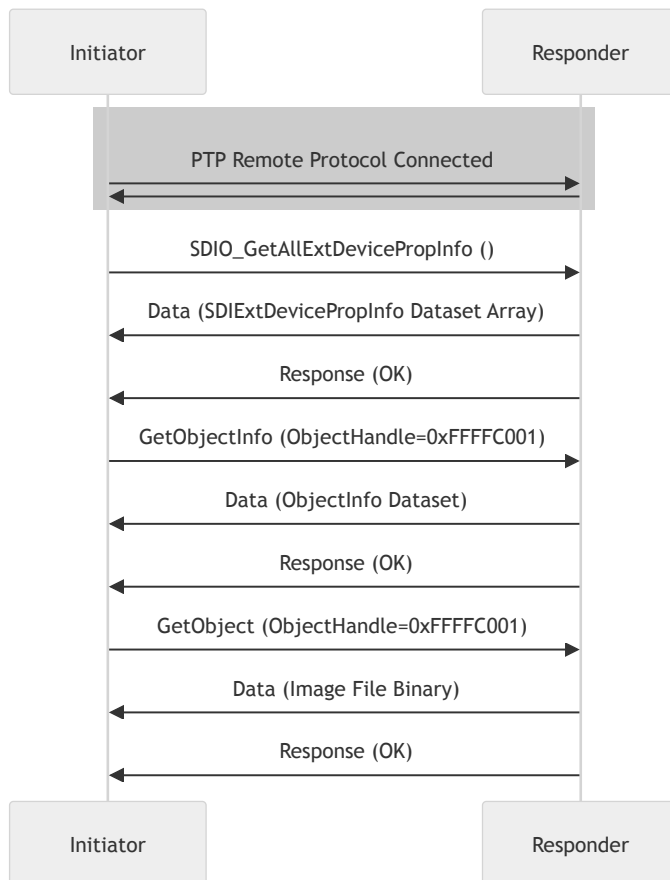
ensured. By contrast, a one-shot release command can be used independently. Therefore, that if the camera is in an acceptable mode (single-shot drive mode), the Initiator can send a one-shot release command anytime.

Note: The Initiator cannot obtain the shooting results. If the device property [Shooting File Info](#) is not zero, and the most significant bit (**0x8000**) is set, the camera shoots, and the image is stored in the buffer.

The Initiator can send a differential for up/down controls such as the F-number, shutter speed, ISO, exposure compensation, and flash compensation. For example, a differential is one step plus/minus or two steps plus/minus. The results of the change will only be sent to the Initiator if it queries the device property values again.

Get Image File

Get a File Sequence:



The Initiator uses the [GetObject](#) command to obtain the image file from the buffer. To obtain the image in the buffer, use **0xFFFFC001** as the object handle. This fixed value always indicates the first file to be transferred. The Initiator should get the same object handle repeatedly until the value of the device property [Shooting File Info](#) becomes zero.

[GetObjectInfo](#) (a PTP standard command) is also available. The Initiator can understand the file type based on the object-format value in the `ObjectInfo` structure.

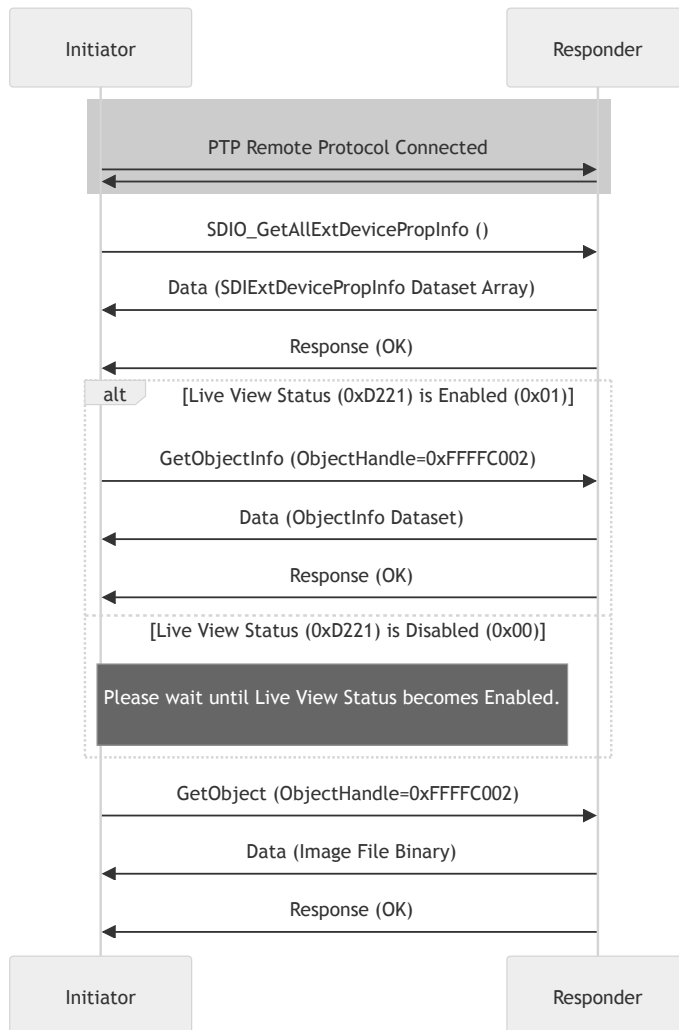
In the DCF (Exif) specification, if the image file has “AdobeRGB” color space, the image file name should start with “_” (e.g., `_DSC0000.JPG`). However, the Initiator cannot obtain the color space information from [GetObjectInfo](#). The Initiator then needs to parse the Exif tags in the image.

When the Initiator sets “RAW+JPEG” or “RAW+HEIF” as the file format, you might want to distinguish the pair of RAW and JPEG or RAW and HEIF. To do this, check the sequence number in the `ObjectInfo` structure that the Initiator can obtain using [GetObjectInfo](#). The file order is JPEG/HEIF, then RAW. Therefore, if the image the Initiator receives is a RAW file, the previous image is a JPEG or HEIF file, and the sequence numbers are the same, it would be the RAW and JPEG/HEIF pair. But the sequence number starts from the same value each time you turn on the camera; if you shoot one JPEG/HEIF via remote and turn off and on the camera, then you shoot one RAW, those JPEG/HEIF and RAW would have the same sequence number. Checking the Exif date time is necessary to distinguish the pair.

After shooting a still image, it must get an image like the “Get a File Sequence” above. It is necessary to obtain the device property [Shooting File Info](#) to confirm the number of images before content transfer. Images exist: [GetObjectInfo](#) and [GetObject](#) must be executed repeatedly the number of images times. No image: [GetObjectInfo](#) and [GetObject](#) need not be executed.

Live View

Live View Sequence:



The Initiator can obtain live-view images in the shooting mode. Refer to [GetObject](#) for the live view.

Movie Recording

Recorded movie files are stored in the memory card in the camera.

Disconnect

The Initiator must execute [CloseSession](#) to disconnect. Subsequently, release the PTP connection.

Command List

The following described a list of operation, device property, control, event, etc.

Operations

Name	Code	Description
GetDeviceInfo	0x1001	same as PIMA 15740
OpenSession	0x1002	same as PIMA 15740
CloseSession	0x1003	same as PIMA 15740
GetObject	0x1009	same as PIMA 15740
GetThumb	0x100A	same as PIMA 15740
GetPartialObject	0x101B	same as PIMA 15740
SDIO_Connect	0x9201	This is for the authentication handshake.
SDIO_GetExtDeviceInfo	0x9202	Get the protocol version and the supported properties of the connected device.
SDIO_SetExtDevicePropValue	0x9205	Set a DevicePropValue for a device property.
SDIO_ControlDevice	0x9207	Set the SDIControl value for the SDIControlCode.
SDIO_GetAllExtDevicePropInfo	0x9209	Obtain all support DevicePropDescs at one time. The host will send this operation at regular intervals to obtain the latest (current) camera settings.

Device Properties

Name	Code	Description
Compression Setting	0x5004	Get/Set the quality.
White Balance	0x5005	Get/Set the white balance.
F-Number	0x5007	Get/Set the aperture value.
Focus Mode	0x500A	Get/Set the focus mode.
Exposure Metering Mode	0x500B	Get/Set the exposure metering mode.
Flash Mode	0x500C	Get/Set the flash mode.
Exposure Program Mode	0x500E	Get/Set the exposure program mode.
Exposure Bias Compensation	0x5010	Get/Set the exposure bias compensation.
Still Capture Mode	0x5013	Get/Set the drive mode.
Flash Compensation	0xD200	Get/Set the flash compensation.
Dynamic Range Optimizer	0xD201	Get/Set the dynamic range optimizer.

Name	Code	Description
Image Size	0xD203	Get/Set the image size.
Shutter Speed	0xD20D	Get/Set the shutter speed.
Battery Level Indicator	0xD20E	Get the battery level indicator.
Color Temperature	0xD20F	Get/Set the color temperature.
Biaxial Fine-Tuning G-M Direction	0xD210	Get/Set the biaxial fine-tuning G-M direction.
Aspect Ratio	0xD211	Get/Set the aspect ratio.
Focus Indication	0xD213	Get the focus indication.
Predicted Maximum File Size	0xD214	Get the predicted maximum file size.
Shooting File Info	0xD215	Get the shooting file info.
AELock Indication	0xD217	Get the AE lock indication.
Battery Remaining	0xD218	Get the battery remaining (%).
Picture Effect	0xD21B	Get/Set the picture effect value.
Biaxial Fine-Tuning A-B Direction	0xD21C	Get/Set the biaxial fine-tuning A-B direction.
Movie Recording State	0xD21D	Get the movie recording state.
ISO Sensitivity	0xD21E	Get/Set the ISO sensitivity.
FELock Indication	0xD21F	Get the FE lock indication.
Live View Status	0xD221	Get the live view status.
Still Image Save Destination	0xD222	Get the information of still image save destination.

Controls

Name	Code	Description
Shutter Half-Release (S1) Button	0xD2C1	Control shutter half-release (S1) button.
Shutter Release (S2) Button	0xD2C2	Control shutter release (S2) button.
AEL Button	0xD2C3	Control AEL button.
AFL Button	0xD2C4	Control AFL button.
Release Lock	0xD2C5	Control release lock.
RequestOneShooting	0xD2C7	Request one shooting.
Movie Rec Button (Hold)	0xD2C8	Control movie recording (S/S) button.
FEL Button	0xD2C9	Control FEL button.
Media Format	0xD2CA	Execute format the media.
Focus Magnifier	0xD2CB	Execute focus magnifier button.
Focus Magnifier Cancel	0xD2CC	Execute focus magnifier cancel button.
Remote Key UP	0xD2CD	Execute remote key UP.
Remote Key DOWN	0xD2CE	Execute remote key DOWN.

Name	Code	Description
Remote Key LEFT	0xD2CF	Execute remote key LEFT.
Remote Key RIGHT	0xD2D0	Execute remote key RIGHT.
AF/MF Hold	0xD2D2	Execute AF/MF hold button.
Cancel Pixel Shift Shooting	0xD2D3	Execute cancel pixel shift shooting button.
Pixel Shift Shooting Mode	0xD2D4	Execute pixel shift shooting mode button.
HFR Standby	0xD2D5	Execute HFR standby.
HFR Recording cancel	0xD2D6	Execute HFR recording cancel.
Focus Step Near	0xD2D7	Execute focus step near.
Focus Step Far	0xD2D8	Execute focus step far.
AWBL Button	0xD2D9	Control AWBL button.

Events

Name	Code	Description
SDIE_ObjectAdded	0xC201	Notify that a shot file is ready to transfer.
SDIE_ObjectRemoved	0xC202	Notify that a shot file is deleted. The device sends this event when the shot file is transferred.
SDIE_DevicePropChanged	0xC203	Notify that the DevicePropValue is changed.

Vendor Response Code

Name	Code	Description
Authentication Failed	0xA101	Indicates that the major version of the connected host is less than that of the camera. It also indicates that authentication fails due to some other factor.
Password Length Over Max	0xA102	Indicates that the password length is over max.
Password Includes Invalid Character	0xA103	Indicates that the password includes invalid character.
Feature Version Invalid Value	0xA104	Indicates that the feature version is invalid value.
Temporary Storage Full	0xA105	Indicates that the temporary storage is full.
Camera Status Error	0xA106	Indicates that a camera status error occurred.

Object Format

Name	Code	Description
EXIF/JPG	0x3801	For JPEG File

Name	Code	Description
RAW	0xB101	For RAW File
HEIF	0xB110	For HEIF File
MPO	0xB301	For MPO File
JFIF	0x3808	For JFIF File

Compatibility

Operations

Code	Name	ILCE-1	ILCE-9M2	ILCE-7RM5	ILCE-7RM4	ILCE-7M4	ILCE-7SM3	ILCE-7RM3	ILCE-7M3	ILCE-7M2	ILCE-7CM2	ILCE-7CR	ILCE-7C	ILCE-6700	ILCE-6600	ILCE-6400	ILCE-6100	ILX-LR1	ILME-FX3	ILME-FX30	ZV-E1	ZV-E10	ZV-1M2	ZV-1F	ZV-1	DSC-RX0M2	DSC-RX100M7	DSC-RX100M5	DSC-RX100M4	DSC-HX99
0x1001	GetDeviceInfo	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0x1002	OpenSession	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0x1003	CloseSession	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0x1009	GetObject	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0x100A	GetThumb	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0x101B	GetPartialObject	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0x9201	SDIO_Connect	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0x9202	SDIO_GetExtDeviceInfo	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0x9205	SDIO_SetExtDevicePropValue	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0x9207	SDIO_ControlDevice	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0x9209	SDIO_GetAllExtDevicePropInfo	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Device Properties

Code	Name	ILCE-1	ILCE-9M2	ILCE-7RM5	ILCE-7RM4	ILCE-7M4	ILCE-7SM3	ILCE-7RM3	ILCE-7M3	ILCE-7M2	ILCE-7CM2	ILCE-7CR	ILCE-7C	ILCE-6700	ILCE-6600	ILCE-6400	ILCE-6100	ILX-LR1	ILME-FX3	ILME-FX30	ZV-E1	ZV-E10	ZV-1M2	ZV-1F	ZV-1	DSC-RX0M2	DSC-RX100M7	DSC-RX100M5	DSC-RX100M4	DSC-HX99
0x5004	Compression Setting	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0x5005	White Balance	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0x5007	F-Number	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0x500A	Focus Mode	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0x500B	Exposure Metering Mode	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0x500C	Flash Mode	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0x500E	Exposure Program Mode	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0x5010	Exposure Bias Compensation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0x5013	Still Capture Mode	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0xD200	Flash Compensation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0xD201	Dynamic Range Optimizer	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0xD203	Image Size	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0xD20D	Shutter Speed	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0xD20E	Battery Level Indicator	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0xD20F	Color Temperature	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0xD210	Biaxial Fine-Tuning G-M Direction	✓	✓		✓		✓	✓	✓				✓		✓	✓	✓					✓			✓	✓	✓	✓	✓	✓
0xD211	Aspect Ratio	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0xD213	Focus Indication	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0xD214	Predicted Maximum File Size	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0xD215	Shooting File Info	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0xD217	AELock Indication	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0xD218	Battery Remaining	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0xD21B	Picture Effect	✓	✓		✓		✓	✓	✓	✓			✓		✓	✓	✓					✓			✓	✓	✓	✓	✓	✓

Code	Name	ILCE-1	ILCE-9M2	ILCE-7RM5	ILCE-7RM4	ILCE-7M4	ILCE-7SM3	ILCE-7RM3	ILCE-7M3	ILCE-7M2	ILCE-7CM2	ILCE-7CR	ILCE-7C	ILCE-6700	ILCE-6600	ILCE-6400	ILCE-6100	ILX-LR1	ILME-FX3	ILME-FX30	ZV-E1	ZV-E10	ZV-1M2	ZV-1F	ZV-1	DSC-RX0M2	DSC-RX100M7	DSC-RX100M5	DSC-RX10M4	DSC-HX99
0xD21C	Biaxial Fine-Tuning A-B Direction	✓	✓		✓		✓	✓	✓	✓			✓		✓	✓	✓	✓				✓			✓	✓	✓	✓	✓	✓
0xD21D	Movie Recording State	✓	✓		✓		✓	✓	✓	✓			✓		✓	✓	✓	✓				✓			✓	✓	✓	✓	✓	✓
0xD21E	ISO Sensitivity	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓						
0xD21F	FELock Indication	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓						
0xD221	Live View Status	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0xD222	Still Image Save Destination	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓				

Controls

Code	Name	ILCE-1	ILCE-9M2	ILCE-7RM5	ILCE-7RM4	ILCE-7M4	ILCE-7SM3	ILCE-7RM3	ILCE-7M3	ILCE-7M2	ILCE-7CM2	ILCE-7CR	ILCE-7C	ILCE-6700	ILCE-6600	ILCE-6400	ILCE-6100	ILX-LR1	ILME-FX3	ILME-FX30	ZV-E1	ZV-E10	ZV-1M2	ZV-1F	ZV-1	DSC-RX0M2	DSC-RX100M7	DSC-RX100M5	DSC-RX10M4	DSC-HX99
0xD2C1	Shutter Half-Release (S1) Button	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0xD2C2	Shutter Release (S2) Button	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0xD2C3	AEL Button	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0xD2C4	AFL Button			✓		✓				✓	✓			✓				✓	✓	✓	✓		✓	✓						
0xD2C5	Release Lock	✓	✓		✓		✓	✓	✓	✓			✓		✓	✓	✓					✓			✓	✓	✓	✓	✓	✓
0xD2C7	RequestOneShooting	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0xD2C8	Movie Rec Button (Hold)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0xD2C9	FEL Button	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			
0xD2CA	Media Format				✓																	✓			✓	✓				
0xD2CB	Focus Magnifier							✓	✓						✓	✓	✓													✓
0xD2CC	Focus Magnifier Cancel							✓	✓						✓	✓	✓													✓
0xD2CD	Remote Key UP			✓		✓		✓	✓	✓	✓	✓		✓	✓	✓	✓	✓	✓	✓	✓		✓	✓						✓
0xD2CE	Remote Key DOWN			✓		✓		✓	✓	✓	✓	✓		✓	✓	✓	✓	✓	✓	✓	✓		✓	✓						✓
0xD2CF	Remote Key LEFT			✓		✓		✓	✓	✓	✓	✓		✓	✓	✓	✓	✓	✓	✓	✓		✓	✓						✓
0xD2D0	Remote Key RIGHT			✓		✓		✓	✓	✓	✓	✓		✓	✓	✓	✓	✓	✓	✓	✓		✓	✓						✓
0xD2D2	AF/MF Hold							✓	✓						✓	✓	✓													✓
0xD2D3	Cancel Pixel Shift Shooting							✓	✓						✓	✓	✓													✓
0xD2D4	Pixel Shift Shooting Mode							✓	✓						✓	✓	✓													✓
0xD2D5	HFR Standby	✓	✓		✓		✓						✓									✓			✓	✓	✓			
0xD2D6	HFR Recording cancel	✓	✓		✓		✓						✓									✓			✓	✓	✓			
0xD2D7	Focus Step Near	✓	✓		✓		✓						✓									✓			✓	✓	✓			
0xD2D8	Focus Step Far	✓	✓		✓		✓						✓									✓			✓	✓	✓			
0xD2D9	AWBL Button	✓	✓		✓		✓						✓		✓	✓	✓	✓	✓	✓	✓	✓			✓	✓	✓			✓

Events

Code	Name	ILCE-1	ILCE-9M2	ILCE-7RM5	ILCE-7RM4	ILCE-7M4	ILCE-7SM3	ILCE-7RM3	ILCE-7M3	ILCE-7M2	ILCE-7CM2	ILCE-7CR	ILCE-7C	ILCE-6700	ILCE-6600	ILCE-6400	ILCE-6100	ILX-LR1	ILME-FX3	ILME-FX30	ZV-E1	ZV-E10	ZV-1M2	ZV-1F	ZV-1	DSC-RX0M2	DSC-RX100M7	DSC-RX100M5	DSC-RX10M4	DSC-HX99
0xC201	SDIE_ObjectAdded	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0xC202	SDIE_ObjectRemoved	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
0xC203	SDIE_DevicePropChanged	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Operations

This chapter provides the detailed operation specification of Camera Control PTP.

GetDeviceInfo

Summary

Same as PIMA 15740 (operation code is `0x1001`.)

Description

same as PIMA 15740

OpenSession

Summary

Same as PIMA 15740 (operation code is `0x1002`.)

Description

same as PIMA 15740

CloseSession

Summary

Same as PIMA 15740 (operation code is `0x1003`.)

Description

same as PIMA 15740

GetObject

Summary

Same as PIMA 15740 (operation code is `0x1009`.)

Description

same as PIMA 15740

Parameters

ObjectHandle

- Use the ObjectHandle `0xFFFFC001` to obtain a captured image.
- Use the ObjectHandle `0xFFFFC002` to obtain a live-view image.

- If the current value of the device property **Live View Status** is enabled, the Initiator can get the following live-view dataset.
- If the current value of the device property **Live View Status** is not enabled, the Responder will stall and return the response Invalid ObjectHandle (0x2009).

LiveView Dataset (ObjectHandle 0xFFFFC002)

Field	Field Order	Size (Bytes)	Datatype
Offset to Live View Image	1	4	UNIT32
Live View Image Size	2	4	UINT32
Reserved	3	Variable	N/A
Live View Image	4	Live View Image Size	JPEG Data

Note

The maximum number of live view fps (frames per second) is 30. Therefore, the Initiator should not use this operation over two times within 33 msec. If the Initiator gets the offset to the live-view image, the live-view image size will be zero in the data phase and **Access_Denied** (0x200F) in the response phase. The host should retry this command to obtain the next live view image.

GetThumb

Summary

Same as PIMA 15740 (operation code is 0x100A.)

Description

same as PIMA 15740

Responses

The camera returns Thumbnail_Not_Present when the Host sends this command because the camera does not create any thumbnails for the remote protocol.

Same as that of PIMA 15740 for other errors.

It makes no sense to send this command.

GetPartialObject

Summary

Same as PIMA 15740 (operation code is 0x101B.)

Description

same as PIMA 15740

Note

It is highly recommended to use GetObject instead of this command for transfer performance.

SDIO_Connect

Summary

This is for the authentication handshake.

Description

Field	Value
Operation Code	0x9201
Operation Parameter 1	Phase Type
Operation Parameter 2	KeyCode1
Operation Parameter 3	KeyCode2
Operation Parameter 4	None
Operation Parameter 5	None
Data	UINT64 Value
Data Direction	R->I
Response Code	OK, Operation_Not_Supported, Session_Not_Open, Invalid_TransactionID, Device_Busy, Parameter_Not_Supported, Authentication_Failed
Response Parameter 1	None
Response Parameter 2	None
Response Parameter 3	None
Response Parameter 4	None
Response Parameter 5	None

Note

The Initiator should send this command with both KeyCode1 and KeyCode2, which are 0x00000000.

SDIO_GetExtDeviceInfo

Summary

Get the protocol version and the supported properties of the connected device.

Description

Field	Value
Operation Code	0x9202
Operation Parameter 1	Initiator Version
Operation Parameter 2	None
Operation Parameter 3	None
Operation Parameter 4	None
Operation Parameter 5	None
Data	SDIExtDeviceInfo Dataset
Data Direction	R->I
Response Code	OK, Operation_Not_Supported, Session_Not_Open, Invalid_TransactionID, Device_Busy, Parameter_Not_Supported, Authentication_Failed
Response Parameter 1	None
Response Parameter 2	None
Response Parameter 3	None
Response Parameter 4	None
Response Parameter 5	None

Data

SDIExtDeviceInfo Dataset

Field	Field Order	Size (Bytes)	Datatype
SDIExtensionVersion	1	2	UINT16
SDIDevicePropCode Array	2	Variable	UINT16 Array
SDIControlCode Array	3	Variable	UINT16 Array

Note

The Initiator Version and SDIExtensionVersion are 0x00C8 (2.00) in this version. 2 indicates the major version, and .00 indicates the minor version.

The camera returns Authentication_Failed if the major version of the connected Initiator is less than the major version.

SDIO_SetExtDevicePropValue

Summary

Set a DevicePropValue for a device property.

Description

Field	Value
Operation Code	0x9205
Operation Parameter 1	DevicePropCode
Operation Parameter 2	None
Operation Parameter 3	None
Operation Parameter 4	None
Operation Parameter 5	None
Data	DevicePropValue
Data Direction	I->R
Response Code	OK, Operation_Not_Supported, Session_Not_Open, Invalid_TransactionID, Device_Busy, Parameter_Not_Supported
Response Parameter 1	None
Response Parameter 2	None
Response Parameter 3	None
Response Parameter 4	None
Response Parameter 5	None

Note

Refer to the [Device Properties](#) for details.

SDIO_ControlDevice

Summary

Set the SDIControl value for the SDIControlCode.

Description

Field	Value
Operation Code	0x9207
Operation Parameter 1	SDIControlCode
Operation Parameter 2	None
Operation Parameter 3	None
Operation Parameter 4	None
Operation Parameter 5	None
Data	SDIControl Value
Data Direction	I->R
Response Code	OK, Operation_Not_Supported, Session_Not_Open, Invalid_TransactionID, Device_Busy, Parameter_Not_Supported
Response Parameter 1	None
Response Parameter 2	None
Response Parameter 3	None
Response Parameter 4	None
Response Parameter 5	None

Note

The camera returns OK, except when the PTP/MTP Protocol fails. For example, when the Initiator sends S2 ON but the camera fails to release a shutter, the response of this command will be OK.

Refer to the [Controls](#) for details.

SDIO_GetAllExtDevicePropInfo

Summary

Obtain all support DevicePropDescs at one time.

The host will send this operation at regular intervals to obtain the latest (current) camera settings.

Description

Field	Value
Operation Code	0x9209
Operation Parameter 1	None
Operation Parameter 2	None
Operation Parameter 3	None
Operation Parameter 4	None
Operation Parameter 5	None
Data	SDIDevicePropInfo Dataset Array
Data Direction	R->I
Response Code	OK, Operation_Not_Supported, Session_Not_Open, Invalid_TransactionID, Device_Busy, Parameter_Not_Supported
Response Parameter 1	None
Response Parameter 2	None
Response Parameter 3	None
Response Parameter 4	None
Response Parameter 5	None

Data

SDIExtDevicePropInfo Dataset

Field	Field Order	Size (Bytes)	Datatype
Device Property Code	1	2	UINT16
DataType	2	2	UINT16
GetSet	3	1	UINT8
IsEnabled	4	1	UINT8
Factory Default Value	5	Variable	Any
Current Value	6	Variable	Any
Form Flag	7	1	UINT8

The GetSet field is defined as the following:

- 0x00: The Initiator cannot set the value
- 0x01: The Initiator can set the value

The IsEnabled field is defined as the following (for the Initiator UI):

- 0x00: False (means invalid, greyed-out, no indication for the button, combo box, some values like shutter speed, F-number)
- 0x01: True (means valid, indication for the button, combo box, some values like shutter speed, F-number)
- 0x02: DispOnly (means only indication; cannot change the value)

If IsEnabled is “False”, the Factory Default Value and Current Value are not guaranteed.

If Form Flag is 0x02 (Enumeration), the Dataset will be the following example:

Field	Field Order	Size (Bytes)	Datatype
Device Property Code	1	2	UINT16
DataType	2	2	UINT16
GetSet	3	1	UINT8
IsEnabled	4	1	UINT8
Factory Default Value	5	Variable	Any
Current Value	6	Variable	Any
Form Flag	7	1	UINT8
Num of Enum lists (Set)	8	2	UINT16
Enum value (Set)[0]	9	Depends on DataType	-
...	...	Depends on DataType	-
Enum value (Set) [Num of Enum lists (Set) - 1]	N	Depends on DataType	-
Num of Enum lists(Get/Set)	N+1	2	UINT16
Enum value (Get/Set)[0]	N+2	Depends on DataType	-
...	...	Depends on DataType	-
Enum value (Get/Set) [Num of Enum lists (Get/Set) - 1]	O	Depends on DataType	-

SDDevicePropInfo Dataset Array

Field	Field Order	Size (Bytes)	Datatype
Num of Elements	1	8	UINT64
SDDevicePropInfo Dataset[0]	2	variable	UINT16
...	...	variable	UINT8
SDDevicePropInfo Dataset [Num of Elements - 1]	N	variable	UINT8

Device Properties

This chapter provides the detailed device property specification of Camera Control PTP.

Compression Setting

Summary

Get/Set the quality.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
PropertyCode	1	2	UINT16	0x5004
Datatype	2	2	UINT16	0x0002(UINT8)
Get/Set	3	1	UINT8	0x01(Get/Set)
IsEnabled	4	1	UINT8	variable
Reserved	5	1	UINT8	0x00
CurrentValue	6	1	UINT8	variable
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x01	ECO/LIGHT
0x02	STD
0x03	FINE
0x04	XFINE
0x10	RAW
0x11	RAW+JPG(LIGHT)
0x12	RAW+JPG(STD)
0x13	RAW+JPG(FINE)
0x14	RAW+JPG(XFINE)
0x20	RAW(compression)
0x23	RAW(compression)+JPG(FINE)
0x31	HEIF(ECO/LIGHT)
0x32	HEIF(STD)
0x33	HEIF(FINE)
0x34	HEIF(XFINE)

Value	Description
0x41	RAW+HEIF(LIGHT)
0x42	RAW+HEIF(STD)
0x43	RAW+HEIF(FINE)
0x44	RAW+HEIF(XFINE)

White Balance

Summary

Get/Set the white balance.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
PropertyCode	1	2	UINT16	0x5005
Datatype	2	2	UINT16	0x0004(UINT16)
Get/Set	3	1	UINT8	0x01(Get/Set)
IsEnabled	4	1	UINT8	variable
Reserved	5	2	UINT16	0x0000
CurrentValue	6	2	UINT16	variable
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0001	Manual
0x0002	AWB
0x0003	One-push Automatic
0x0004	Daylight
0x0005	Fluorescent
0x0006	Tungsten (Incandescent)
0x0007	Flash
0x8001	Fluor.: Warm White (-1)
0x8002	Fluor.: Cool White (0)
0x8003	Fluor.: Day White (+1)
0x8004	Fluor.: Daylight White (+2)
0x8010	Cloudy
0x8011	Shade
0x8012	C.Temp.

Value	Description
0x8020	Custom 1
0x8021	Custom 2
0x8022	Custom 3
0x8023	Custom
0x8030	Underwater Auto

F-Number

Summary

Get/Set the aperture value.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
PropertyCode	1	2	UINT16	0x5007
Datatype	2	2	UINT16	0x0004(UINT16)
Get/Set	3	1	UINT8	0x01(Get/Set)
IsEnabled	4	1	UINT8	variable
Reserved	5	2	UINT16	0xFFFF
CurrentValue	6	2	UINT16	variable
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0064	1
0x006E	1.1
0x0078	1.2
0x0082	1.3
0x008C	1.4
0x00A0	1.6
0x00AA	1.7
0x00B4	1.8
0x00C8	2
0x00DC	2.2
0x00F0	2.4
0x00FA	2.5
0x0118	2.8

Value	Description
0x0136	3.1
0x0140	3.2
0x0154	3.4
0x015E	3.5
0x0172	3.7
0x0190	4
0x01B8	4.4
0x01C2	4.5
0x01E0	4.8
0x01F4	5
0x0208	5.2
0x0230	5.6
0x026C	6.2
0x0276	6.3
0x029E	6.7
0x02A8	6.8
0x02C6	7.1
0x02DA	7.3
0x0320	8
0x0366	8.7
0x0384	9
0x03B6	9.5
0x03C0	9.6
0x03E8	10
0x044C	11
0x0514	13
0x0578	14
0x0640	16
0x0708	18
0x076C	19
0x07D0	20
0x0898	22
0x09C4	25
0x0A8C	27
0x0B54	29

Value	Description
0x0C80	32
0x0E10	36
0x0ED8	38
0x0FA0	40
0x1194	45
0x13EC	51
0x1518	54
0x1644	57
0x1900	64
0x1C20	72
0x1DB0	76
0x1FA4	81
0x2328	90
0xFFFD	Iris Close
0xFFFE	--
0xFFFF	nothing to display

Focus Mode

Summary

Get/Set the focus mode.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
PropertyCode	1	2	UINT16	0x500A
Datatype	2	2	UINT16	0x0004(UINT16)
Get/Set	3	1	UINT8	0x01(Get/Set)
IsEnabled	4	1	UINT8	variable
Reserved	5	2	UINT16	0x0000
CurrentValue	6	2	UINT16	variable
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0001	Manual(MF)
0x0002	Automatic(AF_S)

Value	Description
0x0003	Automatic Macro(close-up)
0x8004	Continuous AF(AF_C)
0x8005	Auto(AF_A)
0x8006	Direct Manual Focus(DMF)
0x8007	Manual Focus Reverse(MF_R)
0x8008	(AF-D)
0x8009	Preset Focus(PF)

Exposure Metering Mode

Summary

Get/Set the exposure metering mode.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
PropertyCode	1	2	UINT16	0x500B
Datatype	2	2	UINT16	0x0004(UINT16)
Get/Set	3	1	UINT8	0x01(Get/Set)
IsEnabled	4	1	UINT8	variable
Reserved	5	2	UINT16	0x0000
CurrentValue	6	2	UINT16	variable
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0001	Average
0x0002	Center-weighted-average
0x0003	Multi-spot
0x0004	Center-spot
0x8001	Multi
0x8002	Center-weighted
0x8003	Entire Screen Avg.
0x8004	Spot : Standard
0x8005	Spot : Large
0x8006	Highlight
0x8011	Standard

Value	Description
0x8012	Backlight
0x8013	Spotlight

Flash Mode

Summary

Get/Set the flash mode.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
PropertyCode	1	2	UINT16	0x500C
Datatype	2	2	UINT16	0x0004(UINT16)
Get/Set	3	1	UINT8	0x01(Get/Set)
IsEnabled	4	1	UINT8	variable
Reserved	5	2	UINT16	0x0000
CurrentValue	6	2	UINT16	variable
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0001	Auto flash
0x0002	Flash off
0x0003	Fill flash
0x0004	Red eye auto
0x0005	Red eye fill
0x0006	External Sync
0x8001	Slow Sync
0x8003	Rear Sync
0x8004	Wireless
0x8021	HSS auto
0x8022	HSS fill
0x8024	HSS WL
0x8031	Slow Sync Red Eye On
0x8032	Slow Sync Red Eye Off
0x8041	Slow Sync Wireless
0x8042	Rear Sync Wireless

Exposure Program Mode

Summary

Get/Set the exposure program mode.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
PropertyCode	1	2	UINT16	0x500E
Datatype	2	2	UINT16	0x0004(UINT16)
Get/Set	3	1	UINT8	0x01(Get/Set)
IsEnabled	4	1	UINT8	variable
Reserved	5	2	UINT16	0x0000
CurrentValue	6	2	UINT16	variable
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0001	Manual (M)
0x0002	Automatic (P)
0x0003	Aperture Priority (A)
0x0004	Shutter Priority (S)
0x0005	Program Creative (Greater Depth of Field)
0x0006	Program Action (Faster Shutter Speed)
0x0007	Portrait
0x8000	Auto
0x8001	Auto+
0x8008	P_A
0x8009	P_S
0x8011	Sports Action
0x8012	Sunset
0x8013	Night Scene
0x8014	Landscape
0x8015	Macro
0x8016	Hand-held Twilight
0x8017	Night Portrait
0x8018	Anti Motion Blur
0x8019	Pet

Value	Description
0x801A	Gourmet
0x801B	Fireworks
0x801C	High Sensitivity
0x8020	Memory Recall (MR)
0x8030	Continuous Priority AE
0x8031	Tele-Zoom Continuous Priority AE 8pics
0x8032	Tele-Zoom Continuous Priority AE 10pics
0x8033	Continuous Priority AE12pics
0x8040	3D Sweep Panorama Shooting
0x8041	Sweep Panorama Shooting
0x8050	Movie Recording (P)
0x8051	Movie Recording (A)
0x8052	Movie Recording (S)
0x8053	Movie Recording (M)
0x8054	Movie Recording (Auto)
0x8059	Movie Recording (S&Q Motion (P))
0x805A	Movie Recording (S&Q Motion (A))
0x805B	Movie Recording (S&Q Motion (S))
0x805C	Movie Recording (S&Q Motion (M))
0x805D	Movie Recording (S&Q Motion (Auto))
0x8060	Flash Off
0x8070	Picture Effect
0x8080	High Frame Rate (P)
0x8081	High Frame Rate (A)
0x8082	High Frame Rate (S)
0x8083	High Frame Rate (M)
0x8084	S&Q Motion (P)
0x8085	S&Q Motion (A)
0x8086	S&Q Motion (S)
0x8087	S&Q Motion (M)
0x8088	Movie
0x8089	Still
0x808A	F (Movie or S&Q)
0x8090	Movie F Mode
0x8091	S&Q F Mode

Value	Description
0x8092	Interval REC (Movie) F Mode
0x8093	Interval REC (Movie) (P)
0x8094	Interval REC (Movie) (A)
0x8095	Interval REC (Movie) (S)
0x8096	Interval REC (Movie) (M)
0x8097	Interval REC (Movie) (Auto)

Exposure Bias Compensation

Summary

Get/Set the exposure bias compensation.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
PropertyCode	1	2	UINT16	0x5010
Datatype	2	2	UINT16	0x0003(INT16)
Get/Set	3	1	UINT8	0x01(Get/Set)
IsEnabled	4	1	UINT8	variable
Reserved	5	2	UINT16	0x0000
CurrentValue	6	2	UINT16	variable
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
Any	The value is obtained by multiplying a real Exposure Bias Compensation value by 1000. e.g.) 0xEC78 = -5000 (means -5.0Ev) 0x0000 = 0 (means 0.0Ev) 0x1388 = 5000 (means 5.0Ev)

Still Capture Mode

Summary

Get/Set the drive mode.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
PropertyCode	1	2	UINT16	0x5013
Datatype	2	2	UINT16	0x0004(UINT16)

Field	Field Order	Size (Bytes)	Datatype	Value
Get/Set	3	1	UINT8	0x01(Get/Set)
IsEnabled	4	1	UINT8	variable
Reserved	5	2	UINT16	0x0000
CurrentValue	6	2	UINT16	variable
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0001	Normal
0x0002	Continuous Shooting Hi
0x8010	Continuous Shooting Hi+
0x8011	Continuous Shooting Hi-Live
0x8012	Continuous Shooting Lo
0x8013	Continuous Shooting
0x8014	Continuous Shooting Speed Priority
0x8015	Continuous Shooting Mid
0x8016	Continuous Shooting Mid-Live
0x8017	Continuous Shooting Lo-Live
0x0003	Timelapse
0x8003	Self Timer 5 Sec.
0x8004	Self Timer 10 Sec.
0x8005	Self Timer 2 Sec.
0xC237	Continuous Bracket 0.3 EV 2 Img. +
0xC23F	Continuous Bracket 0.3 EV 2 Img. -
0x8337	Continuous Bracket 0.3 EV 3 Img.
0x8537	Continuous Bracket 0.3 EV 5 Img.
0x8737	Continuous Bracket 0.3 EV 7 Img.
0x8937	Continuous Bracket 0.3 EV 9 Img.
0xC257	Continuous Bracket 0.5 EV 2 Img. +
0xC25F	Continuous Bracket 0.5 EV 2 Img. -
0x8357	Continuous Bracket 0.5 EV 3 Img.
0x8557	Continuous Bracket 0.5 EV 5 Img.
0x8757	Continuous Bracket 0.5 EV 7 Img.
0x8957	Continuous Bracket 0.5 EV 9 Img.
0xC277	Continuous Bracket 0.7 EV 2 Img. +

Value	Description
0xC27F	Continuous Bracket 0.7 EV 2 Img. -
0x8377	Continuous Bracket 0.7 EV 3 Img.
0x8577	Continuous Bracket 0.7 EV 5 Img.
0x8777	Continuous Bracket 0.7 EV 7 Img.
0x8977	Continuous Bracket 0.7 EV 9 Img.
0xC211	Continuous Bracket 1.0 EV 2 Img. +
0xC219	Continuous Bracket 1.0 EV 2 Img. -
0x8311	Continuous Bracket 1.0 EV 3 Img.
0x8511	Continuous Bracket 1.0 EV 5 Img.
0x8711	Continuous Bracket 1.0 EV 7 Img.
0x8911	Continuous Bracket 1.0 EV 9 Img.
0xC241	Continuous Bracket 1.3 EV 2 Img. +
0xC249	Continuous Bracket 1.3 EV 2 Img. -
0x8341	Continuous Bracket 1.3 EV 3 Img.
0x8541	Continuous Bracket 1.3 EV 5 Img.
0x8741	Continuous Bracket 1.3 EV 7 Img.
0xC261	Continuous Bracket 1.5 EV 2 Img. +
0xC269	Continuous Bracket 1.5 EV 2 Img. -
0x8361	Continuous Bracket 1.5 EV 3 Img.
0x8561	Continuous Bracket 1.5 EV 5 Img.
0x8761	Continuous Bracket 1.5 EV 7 Img.
0xC281	Continuous Bracket 1.7 EV 2 Img. +
0xC289	Continuous Bracket 1.7 EV 2 Img. -
0x8381	Continuous Bracket 1.7 EV 3 Img.
0x8581	Continuous Bracket 1.7 EV 5 Img.
0x8781	Continuous Bracket 1.7 EV 7 Img.
0xC221	Continuous Bracket 2.0 EV 2 Img. +
0xC229	Continuous Bracket 2.0 EV 2 Img. -
0x8321	Continuous Bracket 2.0 EV 3 Img.
0x8521	Continuous Bracket 2.0 EV 5 Img.
0x8721	Continuous Bracket 2.0 EV 7 Img.
0xC251	Continuous Bracket 2.3 EV 2 Img. +
0xC259	Continuous Bracket 2.3 EV 2 Img. -
0x8351	Continuous Bracket 2.3 EV 3 Img.
0x8551	Continuous Bracket 2.3 EV 5 Img.

Value	Description
0xC271	Continuous Bracket 2.5 EV 2 Img. +
0xC279	Continuous Bracket 2.5 EV 2 Img. -
0x8371	Continuous Bracket 2.5 EV 3 Img.
0x8571	Continuous Bracket 2.5 EV 5 Img.
0xC291	Continuous Bracket 2.7 EV 2 Img. +
0xC299	Continuous Bracket 2.7 EV 2 Img. -
0x8391	Continuous Bracket 2.7 EV 3 Img.
0x8591	Continuous Bracket 2.7 EV 5 Img.
0xC231	Continuous Bracket 3.0 EV 2 Img. +
0xC239	Continuous Bracket 3.0 EV 2 Img. -
0x8331	Continuous Bracket 3.0 EV 3 Img.
0x8531	Continuous Bracket 3.0 EV 5 Img.
0xC236	Single Bracket 0.3 EV 2 Img. +
0xC23E	Single Bracket 0.3 EV 2 Img. -
0x8336	Single Bracket 0.3 EV 3 Img.
0x8536	Single Bracket 0.3 EV 5 Img.
0x8736	Single Bracket 0.3 EV 7 Img.
0x8936	Single Bracket 0.3 EV 9 Img.
0xC256	Single Bracket 0.5 EV 2 Img. +
0xC25E	Single Bracket 0.5 EV 2 Img. -
0x8356	Single Bracket 0.5 EV 3 Img.
0x8556	Single Bracket 0.5 EV 5 Img.
0x8756	Single Bracket 0.5 EV 7 Img.
0x8956	Single Bracket 0.5 EV 9 Img.
0xC276	Single Bracket 0.7 EV 2 Img. +
0xC27E	Single Bracket 0.7 EV 2 Img. -
0x8376	Single Bracket 0.7 EV 3 Img.
0x8576	Single Bracket 0.7 EV 5 Img.
0x8776	Single Bracket 0.7 EV 7 Img.
0x8976	Single Bracket 0.7 EV 9 Img.
0xC210	Single Bracket 1.0 EV 2 Img. +
0xC218	Single Bracket 1.0 EV 2 Img. -
0x8310	Single Bracket 1.0 EV 3 Img.
0x8510	Single Bracket 1.0 EV 5 Img.
0x8710	Single Bracket 1.0 EV 7 Img.

Value	Description
0x8910	Single Bracket 1.0 EV 9 Img.
0xC240	Single Bracket 1.3 EV 2 Img. +
0xC248	Single Bracket 1.3 EV 2 Img. -
0x8340	Single Bracket 1.3 EV 3 Img.
0x8540	Single Bracket 1.3 EV 5 Img.
0x8740	Single Bracket 1.3 EV 7 Img.
0xC260	Single Bracket 1.5 EV 2 Img. +
0xC268	Single Bracket 1.5 EV 2 Img. -
0x8360	Single Bracket 1.5 EV 3 Img.
0x8560	Single Bracket 1.5 EV 5 Img.
0x8760	Single Bracket 1.5 EV 7 Img.
0xC280	Single Bracket 1.7 EV 2 Img. +
0xC288	Single Bracket 1.7 EV 2 Img. -
0x8380	Single Bracket 1.7 EV 3 Img.
0x8580	Single Bracket 1.7 EV 5 Img.
0x8780	Single Bracket 1.7 EV 7 Img.
0xC220	Single Bracket 2.0 EV 2 Img. +
0xC228	Single Bracket 2.0 EV 2 Img. -
0x8320	Single Bracket 2.0 EV 3 Img.
0x8520	Single Bracket 2.0 EV 5 Img.
0x8720	Single Bracket 2.0 EV 7 Img.
0xC250	Single Bracket 2.3 EV 2 Img. +
0xC258	Single Bracket 2.3 EV 2 Img. -
0x8350	Single Bracket 2.3 EV 3 Img.
0x8550	Single Bracket 2.3 EV 5 Img.
0xC270	Single Bracket 2.5 EV 2 Img. +
0xC278	Single Bracket 2.5 EV 2 Img. -
0x8370	Single Bracket 2.5 EV 3 Img.
0x8570	Single Bracket 2.5 EV 5 Img.
0xC290	Single Bracket 2.7 EV 2 Img. +
0xC298	Single Bracket 2.7 EV 2 Img. -
0x8390	Single Bracket 2.7 EV 3 Img.
0x8590	Single Bracket 2.7 EV 5 Img.
0xC230	Single Bracket 3.0 EV 2 Img. +
0xC238	Single Bracket 3.0 EV 2 Img. -

Value	Description
0x8330	Single Bracket 3.0 EV 3 Img.
0x8530	Single Bracket 3.0 EV 5 Img.
0x8018	White Balance Bracket Lo
0x8028	White Balance Bracket Hi
0x8019	DRO Bracket Lo
0x8029	DRO Bracket Hi
0x801A	LPF Bracket
0x800A	Remote Commander
0x800B	Mirror Up
0x8006	Self Portrait 1 Person
0x8007	Self Portrait 2 People
0x8008	Continuous Self Timer 3 Img.
0x8009	Continuous Self Timer 5 Img.
0x800C	Continuous Self Timer 3 Img. 5 Sec.
0x800D	Continuous Self Timer 5 Img. 5 Sec.
0x800E	Continuous Self Timer 3 Img. 2 Sec.
0x800F	Continuous Self Timer 5 Img. 2 Sec.
0x8030	Spot Burst Shooting Lo
0x8031	Spot Burst Shooting Mid
0x8032	Spot Burst Shooting Hi
0x8040	Focus Bracket

Flash Compensation

Summary

Get/Set the flash compensation.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
PropertyCode	1	2	UINT16	0xD200
Datatype	2	2	UINT16	0x0003(INT16)
Get/Set	3	1	UINT8	0x01(Get/Set)
IsEnabled	4	1	UINT8	variable
Reserved	5	2	UINT16	0x0000
CurrentValue	6	2	UINT16	variable
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
Any	The value is obtained by multiplying a real Flash Compensation value by 1000. e.g.) 0xEC78 = -5000 (means -5.0Ev) 0x0000 = 0 (means 0.0Ev) 0x1388 = 5000 (means 5.0Ev)

Dynamic Range Optimizer

Summary

Get/Set the dynamic range optimizer.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
PropertyCode	1	2	UINT16	0xD201
Datatype	2	2	UINT16	0x0002(UINT8)
Get/Set	3	1	UINT8	0x01(Get/Set)
IsEnabled	4	1	UINT8	variable
Reserved	5	1	UINT8	0x00
CurrentValue	6	1	UINT8	variable
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x01	DRO OFF
0x02	DRO
0x10	DRO+
0x11	DRO + Manual1
0x12	DRO + Manual2
0x13	DRO + Manual3
0x14	DRO + Manual4
0x15	DRO + Manual5
0x1F	DRO AUTO
0x20	HDR AUTO
0x21	HDR 1.0Ev
0x22	HDR 2.0Ev
0x23	HDR 3.0Ev
0x24	HDR 4.0Ev

Value	Description
0x25	HDR 5.0Ev
0x26	HDR 6.0Ev

Image Size

Summary

Get/Set the image size.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
PropertyCode	1	2	UINT16	0xD203
Datatype	2	2	UINT16	0x0002(UINT8)
Get/Set	3	1	UINT8	0x01(Get/Set)
IsEnabled	4	1	UINT8	variable
Reserved	5	1	UINT8	0x00
CurrentValue	6	1	UINT8	variable
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x01	L
0x02	M
0x03	S
0x04	24M
0x05	16M
0x06	13M
0x07	11M
0x08	8.4M
0x09	6.1M
0x0A	6M
0x0B	5.6M
0x0C	4M
0x0D	2.6M
0x0E	21M
0x0F	20M
0x10	10M

Value	Description
0x11	9.2M
0x12	8.7M
0x13	6.9M
0x14	4.6M
0x15	4.1M
0x16	4M
0x17	3.9M
0x18	3.1M
0x19	VGA

Shutter Speed

Summary

Get/Set the shutter speed.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
PropertyCode	1	2	UINT16	0xD20D
Datatype	2	2	UINT16	0x0006(UINT32)
Get/Set	3	1	UINT8	0x01(Get/Set)
IsEnabled	4	1	UINT8	variable
Reserved	5	4	UINT32	0x00000000
CurrentValue	6	4	UINT32	variable
FormFlag	7	1	UINT8	0x02(Enumlation)

Value

Value	Description
0x00000000	BULB
0xFFFFFFFF	nothing to display
Other than above values	The real value of shutter speed (Upper two bytes: numerator, Lower two bytes: denominator) In the case of the shutter speed is displayed as "Real Number" on the camera, the denominator is fixed 0x000A. e.g.) 0x000F000A: 0x000F (means 15) / 0x000A (means 10) = 1.5" In the case of the shutter speed is displayed as "Fraction Number" on the camera, the numerator is fixed 0x0001. e.g.) 0x000103E8: 0x0001 (means 1) / 0x03E8 (means 1000) = 1/1000

Note

The host should send SDIO_ControlDevice to change this value from the host application

min: 0x00000000

max: 0xFFFFFFFF

step: 0x00000001

Battery Level Indicator

Summary

Get the battery level indicator.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
PropertyCode	1	2	UINT16	0xD20E
Datatype	2	2	UINT16	0x0002(UINT8)
Get/Set	3	1	UINT8	0x00(Get)
IsEnabled	4	1	UINT8	variable
Reserved	5	1	UINT8	0x00
CurrentValue	6	1	UINT8	variable
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x01	Fake Battery
0x02	Unusable
0x03	Pre-End Battery
0x04	Battery Level 1/4
0x05	Battery Level 2/4
0x06	Battery Level 3/4
0x07	Battery Level 4/4
0x08	Battery Level 1/3
0x09	Battery Level 2/3
0x0A	Battery Level 3/3
0x0B	Pre-End Battery with USB Bus Power Supply
0x0C	Battery Level 1/4 with USB Bus Power Supply
0x0D	Battery Level 2/4 with USB Bus Power Supply
0x0E	Battery Level 3/4 with USB Bus Power Supply
0x0F	Battery Level 4/4 with USB Bus Power Supply
0x10	USB Bus Power Supply

Value	Description
0xFF	Battery Not Installed

Color Temperature

Summary

Get/Set the color temperature.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
PropertyCode	1	2	UINT16	0xD20F
Datatype	2	2	UINT16	0x0004(UINT16)
Get/Set	3	1	UINT8	0x01(Get/Set)
IsEnabled	4	1	UINT8	variable
Reserved	5	2	UINT16	0x0000
CurrentValue	6	2	UINT16	variable
FormFlag	7	1	UINT8	0x01(Range)

Value

Value	Description
0x09C4(2500K)	min
0x26AC(9900K)	max
0x0064(100K)	step

Note

The special CurrentValues are the following.

- 0x0000 means less than 2500K.
 - 0xFFFF means greater than 9900K.
- These values are not included the value of Range (they are only used as CurrentValue).

Biaxial Fine-Tuning G-M Direction

Summary

Get/Set the biaxial fine-tuning G-M direction.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
PropertyCode	1	2	UINT16	0xD210
Datatype	2	2	UINT16	0x0002(UINT8)

Field	Field Order	Size (Bytes)	Datatype	Value
Get/Set	3	1	UINT8	0x01(Get/Set)
IsEnabled	4	1	UINT8	variable
Reserved	5	1	UINT8	0x00
CurrentValue	6	1	UINT8	variable
FormFlag	7	1	UINT8	0x01(Range)

Value

Value	Description
0x9C(M9_00)	min
0xE4(G9_00)	max
0x01(0.25)	step

Aspect Ratio

Summary

Get/Set the aspect ratio.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
PropertyCode	1	2	UINT16	0xD211
Datatype	2	2	UINT16	0x0002(UINT8)
Get/Set	3	1	UINT8	0x01(Get/Set)
IsEnabled	4	1	UINT8	variable
Reserved	5	1	UINT8	0x00
CurrentValue	6	1	UINT8	variable
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x01	3:02
0x02	16:09
0x03	4:03
0x04	1:01

Focus Indication

Summary

Get the focus indication.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
PropertyCode	1	2	UINT16	0xD213
Datatype	2	2	UINT16	0x0002(UINT8)
Get/Set	3	1	UINT8	0x00(Get)
IsEnabled	4	1	UINT8	variable
Reserved	5	1	UINT8	0x00
CurrentValue	6	1	UINT8	variable
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x01	Unlock
0x02	●[AF-S]Focused, and AF Locked State
0x03	●[AF-S]Not Focused, and Low Contrast State
0x04	Not Used
0x05	(())[AF-C]Tracking Subject motion
0x06	((●))[AF-C]Focused State
0x07	((●)[AF-C]Not Focused, and Low Contrast State
0x08	Unpause
0x09	Pause

Note

Limitation notice: This device property does not work if HDMI is never outputted on the ILX-LR1 (Ver. 1.00). Please use SDIE_AFStatus in Camera Control PTP 3 instead.

Predicted Maximum File Size

Summary

Get the predicted maximum file size.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
PropertyCode	1	2	UINT16	0xD214

Field	Field Order	Size (Bytes)	Datatype	Value
Datatype	2	2	UINT16	0x0006(UINT32)
Get/Set	3	1	UINT8	0x00(Get)
IsEnabled	4	1	UINT8	variable
Reserved	5	4	UINT32	0x00000000
CurrentValue	6	4	UINT32	variable
FormFlag	7	1	UINT8	0x01(Range)

Value

Value	Description
0x00000000	min
0xFFFFFFFF	max
0x00000001	step

Note

This value is followed by the current settings.

Shooting File Info

Summary

Get the shooting file info.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
PropertyCode	1	2	UINT16	0xD215
Datatype	2	2	UINT16	0x0004(UINT16)
Get/Set	3	1	UINT8	0x00(Get)
IsEnabled	4	1	UINT8	variable
Reserved	5	2	UINT16	0x0000
CurrentValue	6	2	UINT16	variable
FormFlag	7	1	UINT8	0x01(Range)

Value

Value	Description
0x0000	min
0xFFFF	max
0x0001	step

Note

0x0000: Transferable file does not exist

0x0001-0x7FFF: File exists

If the value is over 0x8001 (MSB is 0b01), the host can send GetObjectInfo/GetObject to get the shot files.

AELock Indication

Summary

Get the AE lock indication.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
PropertyCode	1	2	UINT16	0xD217
Datatype	2	2	UINT16	0x0002(UINT8)
Get/Set	3	1	UINT8	0x00(Get)
IsEnabled	4	1	UINT8	variable
Reserved	5	1	UINT8	0x00
CurrentValue	6	1	UINT8	variable
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x01	Unlocked
0x02	Locked

Battery Remaining

Summary

Get the battery remaining (%).

Description

Field	Field Order	Size (Bytes)	Datatype	Value
PropertyCode	1	2	UINT16	0xD218
Datatype	2	2	UINT16	0x0001(INT8)
Get/Set	3	1	UINT8	0x00(Get)
IsEnabled	4	1	UINT8	variable
Reserved	5	1	INT8	0x00
CurrentValue	6	1	INT8	variable
FormFlag	7	1	UINT8	0x01(Range)

Value

Value	Description
0xFF (untaken)	min
0x64 (100%)	max
0x01	step

Picture Effect

Summary

Get/Set the picture effect value.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
PropertyCode	1	2	UINT16	0xD21B
Datatype	2	2	UINT16	0x0004(UINT16)
Get/Set	3	1	UINT8	0x01(Get/Set)
IsEnabled	4	1	UINT8	variable
Reserved	5	2	UINT16	0x0000
CurrentValue	6	2	UINT16	variable
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x8000	OFF
0x8001	Toy Camera Normal
0x8002	Toy Camera Cool
0x8003	Toy Camera Warm
0x8004	Toy Camera Green
0x8005	Toy Camera Magenta
0x8010	Pop Color
0x8020	Posterization B/W
0x8021	Posterization Color
0x8030	Retro Photo
0x8040	Soft High-key
0x8050	Partial Color Red
0x8051	Partial Color Green
0x8052	Partial Color Blue

Value	Description
0x8053	Partial Color Yellow
0x8060	High Contrast Mono
0x8070	Soft Focus Low
0x8071	Soft Focus Mid
0x8072	Soft Focus High
0x8080	HDR Painting Low
0x8081	HDR Painting Mid
0x8082	HDR Painting High
0x8090	Rich-tone Mono
0x80A0	Miniature Auto
0x80A1	Miniature Top
0x80A2	Miniature Middle(Horizontal
0x80A3	Miniature Bottom
0x80A4	Miniature Right
0x80A5	Miniature Middle(Vertical)
0x80A6	Miniature Left
0x80B0	Miniature Water Color
0x80C0	Miniature Illustration Low
0x80C1	Miniature Illustration Mid
0x80C2	Miniature Illustration High

Biaxial Fine-Tuning A-B Direction

Summary

Get/Set the biaxial fine-tuning A-B direction.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
PropertyCode	1	2	UINT16	0xD21C
Datatype	2	2	UINT16	0x0002(UINT8)
Get/Set	3	1	UINT8	0x01(Get/Set)
IsEnabled	4	1	UINT8	variable
Reserved	5	1	UINT8	0x00
CurrentValue	6	1	UINT8	variable
FormFlag	7	1	UINT8	0x01(Range)

Value

Value	Description
0x9C (B9_00)	min
0xE4 (A9_00)	max
0x01 (0.25)	step

Note

The AB value sent to PC App from the camera corresponds to one of the following patterns:
 AB number is BY or AY, where Y is a decimal from 0.00 to 9.00 and increments by 0.25.
 ex) B9.00 (0x9C), B8.75 (0x9D), ..., A8.75 (0xE3), A9.00 (0xE4).

Movie Recording State

Summary

Get the movie recording state.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
PropertyCode	1	2	UINT16	0xD21D
Datatype	2	2	UINT16	0x0002(UINT8)
Get/Set	3	1	UINT8	0x00(Get)
IsEnabled	4	1	UINT8	variable
Reserved	5	1	UINT8	0x00
CurrentValue	6	1	UINT8	variable
FormFlag	7	1	UINT8	0x01(Range)

Value

Value	Description
0x00	Not Recording
0x01	Recording
0x02	Recording Failed
0x03	Waiting Record

Note

“Waiting Record” (0x03) is not record during “Interval REC (movie)” recording.

ISO Sensitivity

Summary

Get/Set the ISO sensitivity.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
PropertyCode	1	2	UINT16	0xD21E
Datatype	2	2	UINT16	0x0006(UINT32)
Get/Set	3	1	UINT8	0x01(Get/Set)
IsEnabled	4	1	UINT8	variable
Reserved	5	2	UINT32	0x00000000
CurrentValue	6	2	UINT32	variable
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0000000A	ISO 10
0x0000000C	ISO 12
0x00000010	ISO 16
0x00000014	ISO 20
0x00000019	ISO 25
0x00000020	ISO 32
0x00000028	ISO 40
0x00000032	ISO 50
0x00000040	ISO 64
0x00000050	ISO 80
0x00000064	ISO 100
0x0000007D	ISO 125
0x000000A0	ISO 160
0x000000C8	ISO 200
0x000000FA	ISO 250
0x00000140	ISO 320
0x00000190	ISO 400
0x000001F4	ISO 500
0x00000280	ISO 640
0x00000320	ISO 800
0x000003E8	ISO 1000
0x000004E2	ISO 1250
0x00000640	ISO 1600
0x000007D0	ISO 2000
0x000009C4	ISO 2500

Value	Description
0x00000C80	ISO 3200
0x00000FA0	ISO 4000
0x00001388	ISO 5000
0x00001900	ISO 6400
0x00001F40	ISO 8000
0x00002710	ISO 10000
0x00003200	ISO 12800
0x00003E80	ISO 16000
0x00004E20	ISO 20000
0x00006400	ISO 25600
0x00007D00	ISO 32000
0x00009C40	ISO 40000
0x0000C800	ISO 51200
0x0000FA00	ISO 64000
0x00013880	ISO 80000
0x00019000	ISO 102400
0x0001F400	ISO 128000
0x00027100	ISO 160000
0x00032000	ISO 204800
0x0003E800	ISO 256000
0x0004E200	ISO 320000
0x00064000	ISO 409600
0x0007D000	ISO 512000
0x0009C400	ISO 640000
0x000C8000	ISO 819200
0x00FFFFFF	ISO AUTO
0x0100000A	Multi Frame NR ISO 10
0x0100000C	Multi Frame NR ISO 12
0x01000010	Multi Frame NR ISO 16
0x01000014	Multi Frame NR ISO 20
0x01000019	Multi Frame NR ISO 25
0x01000020	Multi Frame NR ISO 32
0x01000028	Multi Frame NR ISO 40
0x01000032	Multi Frame NR ISO 50
0x01000040	Multi Frame NR ISO 64

Value	Description
0x01000050	Multi Frame NR ISO 80
0x01000064	Multi Frame NR ISO 100
0x0100007D	Multi Frame NR ISO 125
0x010000A0	Multi Frame NR ISO 160
0x010000C8	Multi Frame NR ISO 200
0x010000FA	Multi Frame NR ISO 250
0x01000140	Multi Frame NR ISO 320
0x01000190	Multi Frame NR ISO 400
0x010001F4	Multi Frame NR ISO 500
0x01000280	Multi Frame NR ISO 640
0x01000320	Multi Frame NR ISO 800
0x010003E8	Multi Frame NR ISO 1000
0x010004E2	Multi Frame NR ISO 1250
0x01000640	Multi Frame NR ISO 1600
0x010007D0	Multi Frame NR ISO 2000
0x010009C4	Multi Frame NR ISO 2500
0x01000C80	Multi Frame NR ISO 3200
0x01000FA0	Multi Frame NR ISO 4000
0x01001388	Multi Frame NR ISO 5000
0x01001900	Multi Frame NR ISO 6400
0x01001F40	Multi Frame NR ISO 8000
0x01002710	Multi Frame NR ISO 10000
0x01003200	Multi Frame NR ISO 12800
0x01003E80	Multi Frame NR ISO 16000
0x01006400	Multi Frame NR ISO 25600
0x0100C800	Multi Frame NR ISO 51200
0x01019000	Multi Frame NR ISO 102400
0x01032000	Multi Frame NR ISO 204800
0x01064000	Multi Frame NR ISO 409600
0x010C8000	Multi Frame NR ISO 819200
0x01FFFFFF	Multi Frame NR ISO AUTO
0x0200000A	Multi Frame NR High ISO 10
0x0200000C	Multi Frame NR High ISO 12
0x02000010	Multi Frame NR High ISO 16
0x02000014	Multi Frame NR High ISO 20

Value	Description
0x02000019	Multi Frame NR High ISO 25
0x02000020	Multi Frame NR High ISO 32
0x02000028	Multi Frame NR High ISO 40
0x02000032	Multi Frame NR High ISO 50
0x02000040	Multi Frame NR High ISO 64
0x02000050	Multi Frame NR High ISO 80
0x02000064	Multi Frame NR High ISO 100
0x0200007D	Multi Frame NR High ISO 125
0x020000A0	Multi Frame NR High ISO 160
0x020000C8	Multi Frame NR High ISO 200
0x020000FA	Multi Frame NR High ISO 250
0x02000140	Multi Frame NR High ISO 320
0x02000190	Multi Frame NR High ISO 400
0x020001F4	Multi Frame NR High ISO 500
0x02000280	Multi Frame NR High ISO 640
0x02000320	Multi Frame NR High ISO 800
0x020003E8	Multi Frame NR High ISO 1000
0x020004E2	Multi Frame NR High ISO 1250
0x02000640	Multi Frame NR High ISO 1600
0x020007D0	Multi Frame NR High ISO 2000
0x020009C4	Multi Frame NR High ISO 2500
0x02000C80	Multi Frame NR High ISO 3200
0x02000FA0	Multi Frame NR High ISO 4000
0x02001388	Multi Frame NR High ISO 5000
0x02001900	Multi Frame NR High ISO 6400
0x02001F40	Multi Frame NR High ISO 8000
0x02002710	Multi Frame NR High ISO 10000
0x02003200	Multi Frame NR High ISO 12800
0x02003E80	Multi Frame NR High ISO 16000
0x02006400	Multi Frame NR High ISO 25600
0x0200C800	Multi Frame NR High ISO 51200
0x02019000	Multi Frame NR High ISO 102400
0x02032000	Multi Frame NR High ISO 204800
0x02064000	Multi Frame NR High ISO 409600
0x020C8000	Multi Frame NR High ISO 819200

Value	Description
0x02FFFFFF	Multi Frame NR High ISO AUTO

Note

To specify extended ISO values, add an offset value 0x10000000.

FE Lock Indication

Summary

Get the FE lock indication.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
PropertyCode	1	2	UINT16	0xD21F
Datatype	2	2	UINT16	0x0002(UINT8)
Get/Set	3	1	UINT8	0x00(Get)
IsEnabled	4	1	UINT8	variable
Reserved	5	1	UINT8	0x00
CurrentValue	6	1	UINT8	variable
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x01	Unlocked
0x02	Locked

Live View Status

Summary

Get the live view status.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
PropertyCode	1	2	UINT16	0xD221
Datatype	2	2	UINT16	0x0002(UINT8)
Get/Set	3	1	UINT8	0x00(Get)
IsEnabled	4	1	UINT8	variable
Reserved	5	1	UINT8	0x00
CurrentValue	6	1	UINT8	variable

Field	Field Order	Size (Bytes)	Datatype	Value
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x00	Live View Supported but Disabled: If this value is set, the host should not get the LiveView Image.
0x01	Live View Supported and Enabled: The host can get the LiveView Image and activate LiveView button if have.
0x02	Live View Not Supported: Just definition, if the camera doesn't support Liveview, the host can't get this property by any operation.

Still Image Save Destination

Summary

Get the information of still image save destination.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
PropertyCode	1	2	UINT16	0xD222
Datatype	2	2	UINT16	0x0004(UINT16)
Get/Set	3	1	UINT8	0x00(Get)
IsEnabled	4	1	UINT8	variable
Reserved	5	1	UINT8	0x0000
CurrentValue	6	1	UINT8	variable
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0001	Host Device (e.g., PC)
0x0010	Camera (Memory Card)
0x0011	Host Device & Camera (Memory Card)

Controls

This chapter provides the detailed control specification of Camera Control PTP.

Shutter Half-Release (S1) Button

Summary

Control shutter half-release (S1) button.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
ControlCode	1	2	UINT16	0xD2C1
Datatype	2	2	UINT16	0x0004(UINT16)
SDIControlType	3	1	UINT8	0x81(Button)
Reserved	4	2	UINT16	0x0000
Reserved	5	2	UINT16	0x0000
FormFlag	6	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0001	Up
0x0002	Down

Shutter Release (S2) Button

Summary

Control shutter release (S2) button.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
ControlCode	1	2	UINT16	0xD2C2
Datatype	2	2	UINT16	0x0004(UINT16)
SDIControlType	3	1	UINT8	0x81(Button)
Reserved	4	2	UINT16	0x0000
Reserved	5	2	UINT16	0x0000
FormFlag	6	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0001	Up
0x0002	Down

AEL Button

Summary

Control AEL button.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
ControlCode	1	2	UINT16	0xD2C3
Datatype	2	2	UINT16	0x0004(UINT16)
SDIControlType	3	1	UINT8	0x81(Button)
Reserved	4	2	UINT16	0x0000
Reserved	5	2	UINT16	0x0000
FormFlag	6	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0001	Up
0x0002	Down

AFL Button

Summary

Control AFL button.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
ControlCode	1	2	UINT16	0xD2C4
Datatype	2	2	UINT16	0x0004(UINT16)
SDIControlType	3	1	UINT8	0x81(Button)
Reserved	4	2	UINT16	0x0000
Reserved	5	2	UINT16	0x0000
FormFlag	6	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0001	Up
0x0002	Down

Release Lock

Summary

Control release lock.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
ControlCode	1	2	UINT16	0xD2C5
Datatype	2	2	UINT16	0x0004(UINT16)
SDIControlType	3	1	UINT8	0x83(Lock)
Reserved	4	2	UINT16	0x0000
Reserved	5	2	UINT16	0x0000
FormFlag	6	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0001	Unlock
0x0002	Lock

RequestOneShooting

Summary

Request one shooting.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
ControlCode	1	2	UINT16	0xD2C7
Datatype	2	2	UINT16	0x0004(UINT16)
SDIControlType	3	1	UINT8	0x81(Button)
Reserved	4	1	UINT8	Variable
Reserved	5	2	UINT16	0x0000
Reserved	6	2	UINT16	0x0000
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0001	Up
0x0002	Down

Movie Rec Button (Hold)

Summary

Control movie recording (S/S) button.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
ControlCode	1	2	UINT16	0xD2C8
Datatype	2	2	UINT16	0x0004(UINT16)
SDIControlType	3	1	UINT8	0x81(Button)
Reserved	4	1	UINT8	Variable
Reserved	5	2	UINT16	0x0000
Reserved	6	2	UINT16	0x0000
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0001	Up
0x0002	Down

Note

Unlike the button action of the main unit, this command functions like holding a button.

FEL Button

Summary

Control FEL button.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
ControlCode	1	2	UINT16	0xD2C9
Datatype	2	2	UINT16	0x0004(UINT16)
SDIControlType	3	1	UINT8	0x81(Button)
Reserved	4	1	UINT8	Variable

Field	Field Order	Size (Bytes)	Datatype	Value
Reserved	5	2	UINT16	0x0000
Reserved	6	2	UINT16	0x0000
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0001	Up
0x0002	Down

Media Format

Summary

Execute format the media.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
ControlCode	1	2	UINT16	0xD2CA
Datatype	2	2	UINT16	0x0004(UINT16)
SDIControlType	3	1	UINT8	0x81(Button)
IsEnabled	4	1	UINT8	variable
Reserved	5	2	UINT16	0x0000
Reserved	6	2	UINT16	0x0000
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0001	Up
0x0002	Down

Note

When executing Media Format (0xD2CA) Down (0x0002), be sure to execute Media Format (0xD61C) Up (0x0001).

Media format is completed when Media Format State (0xD227) has changed to “Not Formatting the Media” (0x00) from “Formatting the Media” (0x01) after executing Media Format (0xD2CA).

Focus Magnifier

Summary

Execute focus magnifier button.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
ControlCode	1	2	UINT16	0xD2CB
Datatype	2	2	UINT16	0x0004(UINT16)
SDIControlType	3	1	UINT8	0x81(Button)
IsEnabled	4	1	UINT8	variable
Reserved	5	2	UINT16	0x0000
Reserved	6	2	UINT16	0x0000
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0001	Up
0x0002	Down

Note

When this Control Code is executed, the following device properties change:

Focus Magnifier Status (0xD22D)

Current Focus Magnifier Ratio (0xD22F)

Focus Magnifier Cancel

Summary

Execute focus magnifier cancel button.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
ControlCode	1	2	UINT16	0xD2CC
Datatype	2	2	UINT16	0x0004(UINT16)
SDIControlType	3	1	UINT8	0x81(Button)
IsEnabled	4	1	UINT8	variable
Reserved	5	2	UINT16	0x0000
Reserved	6	2	UINT16	0x0000
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0001	Up
0x0002	Down

Note

When this Control Code is executed, the following device properties change:

Focus Magnifier Status (0xD22D)

Current Focus Magnifier Ratio (0xD22F)

Remote Key UP

Summary

Execute remote key UP.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
ControlCode	1	2	UINT16	0xD2CD
Datatype	2	2	UINT16	0x0004(UINT16)
SDIControlType	3	1	UINT8	0x81(Button)
Reserved	4	1	UINT8	Variable
Reserved	5	2	UINT16	0x0000
Reserved	6	2	UINT16	0x0000
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0001	Up
0x0002	Down

Note

Execute when the control value is Up (0x0001)

When this Control Code is executed, the following device properties change:

Focus Magnifier Position (x,y) (0xD230)

Remote Key DOWN

Summary

Execute remote key DOWN.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
ControlCode	1	2	UINT16	0xD2CE
Datatype	2	2	UINT16	0x0004(UINT16)
SDIControlType	3	1	UINT8	0x81(Button)
Reserved	4	1	UINT8	Variable
Reserved	5	2	UINT16	0x0000
Reserved	6	2	UINT16	0x0000
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0001	Up
0x0002	Down

Note

Execute when the control value is Up (0x0001)

When this Control Code is executed, the following device properties change:

Focus Magnifier Position (x, y) (0xD230)

Remote Key LEFT**Summary**

Execute remote key LEFT.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
ControlCode	1	2	UINT16	0xD2CF
Datatype	2	2	UINT16	0x0004(UINT16)
SDIControlType	3	1	UINT8	0x81(Button)
Reserved	4	1	UINT8	Variable
Reserved	5	2	UINT16	0x0000
Reserved	6	2	UINT16	0x0000
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0001	Up
0x0002	Down

Note

Execute when the control value is Up (0x0001)

When this Control Code is executed, the following device properties change:

Focus Magnifier Position (x, y) (0xD230)

Remote Key RIGHT

Summary

Execute remote key RIGHT.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
ControlCode	1	2	UINT16	0xD2D0
Datatype	2	2	UINT16	0x0004(UINT16)
SDIControlType	3	1	UINT8	0x81(Button)
Reserved	4	1	UINT8	Variable
Reserved	5	2	UINT16	0x0000
Reserved	6	2	UINT16	0x0000
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0001	Up
0x0002	Down

Note

Execute when the control value is Up (0x0001)

When this Control Code is executed, the following device properties change:

Focus Magnifier Position (x, y) (0xD230)

AF/MF Hold

Summary

Execute AF/MF hold button.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
ControlCode	1	2	UINT16	0xD2D2
Datatype	2	2	UINT16	0x0004(UINT16)
SDIControlType	3	1	UINT8	0x81(Button)

Field	Field Order	Size (Bytes)	Datatype	Value
Reserved	4	1	UINT8	Variable
Reserved	5	2	UINT16	0x0000
Reserved	6	2	UINT16	0x0000
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0001	Up
0x0002	Down

Note

Execute Hold when the control value is Down (0x0002) and cancel Hold when Up (0x0001).

When this Control Code is executed, the following device properties change:

Near/Far Enable Status (0xD235)

Cancel Pixel Shift Shooting

Summary

Execute cancel pixel shift shooting button.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
ControlCode	1	2	UINT16	0xD2D3
Datatype	2	2	UINT16	0x0004(UINT16)
SDIControlType	3	1	UINT8	0x81(Button)
Reserved	4	1	UINT8	Variable
Reserved	5	2	UINT16	0x0000
Reserved	6	2	UINT16	0x0000
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0001	Up
0x0002	Down

Note

Execute when the control value is Up (0x0001)

It is available only when the Pixel Shift Shooting Status (0xD239) of the device property is Burst Shooting (0x01).

When this Control Code is executed, the following device properties change:

Pixel Shift Shooting Mode (0xD239)

Pixel Shift Shooting Mode

Summary

Execute pixel shift shooting mode button.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
ControlCode	1	2	UINT16	0xD2D4
Datatype	2	2	UINT16	0x0004(UINT16)
SDIControlType	3	1	UINT8	0x81(Button)
Reserved	4	1	UINT8	Variable
Reserved	5	2	UINT16	0x0000
Reserved	6	2	UINT16	0x0000
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0001	Off
0x0002	On (Burst Shooting)

Note

Switch setting is made according to the Pixel Shift Shooting Mode (0xD239) of the device property (Off (0x00) / Burst Shooting (0x01)).

When this Control Code is executed, the following device properties change:

Pixel Shift Shooting Mode (0xD239)

HFR Standby

Summary

Execute HFR standby.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
ControlCode	1	2	UINT16	0xD2D5
Datatype	2	2	UINT16	0x0004(UINT16)
SDIControlType	3	1	UINT8	0x81(Button)
Reserved	4	1	UINT8	Variable

Field	Field Order	Size (Bytes)	Datatype	Value
Reserved	5	2	UINT16	0x0000
Reserved	6	2	UINT16	0x0000
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0001	Up
0x0002	Down

Note

When you execute Down (0x0002), be sure to execute Up (0x0001).

HFR Recording cancel

Summary

Execute HFR recording cancel.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
ControlCode	1	2	UINT16	0xD2D6
Datatype	2	2	UINT16	0x0004(UINT16)
SDIControlType	3	1	UINT8	0x81(Button)
Reserved	4	1	UINT8	Variable
Reserved	5	2	UINT16	0x0000
Reserved	6	2	UINT16	0x0000
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0001	Up
0x0002	Down

Note

When you execute Down (0x0002), be sure to execute Up (0x0001).

Focus Step Near

Summary

Execute focus step near.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
ControlCode	1	2	UINT16	0xD2D7
Datatype	2	2	UINT16	0x0004(UINT16)
SDIControlType	3	1	UINT8	0x81(Button)
Reserved	4	1	UINT8	Variable
Reserved	5	2	UINT16	0x0000
Reserved	6	2	UINT16	0x0000
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0001	Up
0x0002	Down

Note

When you execute Down (0x0002), be sure to execute Up (0x0001).

Focus Step Far

Summary

Execute focus step far.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
ControlCode	1	2	UINT16	0xD2D8
Datatype	2	2	UINT16	0x0004(UINT16)
SDIControlType	3	1	UINT8	0x81(Button)
Reserved	4	1	UINT8	Variable
Reserved	5	2	UINT16	0x0000
Reserved	6	2	UINT16	0x0000
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0001	Up
0x0002	Down

Note

When you execute Down (0x0002), be sure to execute Up (0x0001).

AWBL Button

Summary

Control AWBL button.

Description

Field	Field Order	Size (Bytes)	Datatype	Value
ControlCode	1	2	UINT16	0xD2D9
Datatype	2	2	UINT16	0x0004(UINT16)
SDIControlType	3	1	UINT8	0x81(Button)
Reserved	4	1	UINT8	Variable
Reserved	5	2	UINT16	0x0000
Reserved	6	2	UINT16	0x0000
FormFlag	7	1	UINT8	0x02(Enumeration)

Value

Value	Description
0x0001	Up
0x0002	Down

Events

This chapter provides the detailed event specification of Camera Control PTP.

SDIE_ObjectAdded

Summary

Notify that a shot file is ready to transfer.

Description

- Event Code: 0xC201
- Parameter1: ObjectHandle
- Parameter2: None
- Parameter3: None

Note

The host can send GetObjectInfo with the notified ObjectHandle after receiving this event.
The host can choose from checking the value of the Shooting File Info (0xD215) or waiting for this event to send GetObjectInfo.

SDIE_ObjectRemoved

Summary

Notify that a shot file is deleted.
The device sends this event when the shot file is transferred.

Description

- Event Code: 0xC202
- Parameter1: ObjectHandle
- Parameter2: None
- Parameter3: None

Note

The host must not send GetObjectInfo with the notified ObjectHandle after receiving this event.

SDIE_DevicePropChanged

Summary

Notify that the DevicePropValue is changed.

Description

- Event Code: 0xC203
- Parameter1: None

- Parameter2: None
- Parameter3: None

Note

It is highly recommended to use `SDIO_GetAllDevicePropInfo` to obtain current camera-setting values via polling.

More Information

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